

PoMAS Ablation Study - Results So Far

Scope: exploratory seizure-state and preictal classification experiments conducted on CHB-MIT subjects chb01, chb02, chb03, and chb05. This report records observed results; it does not claim clinical performance.

Temporal LSTM baseline

Patient-dependent chb01 exploratory: ROC-AUC 0.725; PR-AUC 0.773; sensitivity 46.0%; specificity 86.7%. Threshold selection was not validation-safe.

Spectral + logistic regression (patient-dependent)

chb01 exploratory: test ROC-AUC 0.899 and PR-AUC 0.921, but validation ROC-AUC was 0.107. The validation-derived threshold produced no useful alerts.

Spectral + logistic regression (patient-independent)

Train chb01+chb02; validate chb03; test chb05: ROC-AUC 0.439; PR-AUC 0.046; preictal sensitivity 5.4%.

SALT LTformer from scratch (patient-independent)

Same multi-subject split: accuracy 78.8%; macro-F1 0.319; preictal recall 2.0%.

SALT LTformer from scratch (patient-dependent)

chb01 held-out seizure recordings: accuracy 45.7%; macro-F1 0.540; preictal recall 6.4%.

Pretrained LaBraM (patient-independent)

Same multi-subject split: accuracy 69.3%; macro-F1 0.278; preictal recall 0.0%.

Interpretation

The shared patient-independent split was train chb01+chb02, validation chb03, test chb05. None of the tested models generalized reliably to the unseen patient chb05. The spectral model, SALT LTformer, and LaBraM all produced low preictal recall or below-chance ranking on this split.

Why accuracy is insufficient

Interictal windows are the dominant class. A model can obtain a high accuracy by predicting mainly interictal, while failing to identify preictal windows. Macro-F1 and preictal recall are therefore the primary indicators in this report.

Ablation Factors and Lessons

	Variants tested	Observed lesson
g regime	Patient-dependent vs patient-independent	Cross-patient generalization was substantially harder than within-patient experiments.
sensation	Raw temporal LSTM, spectral log-power, SALT LTformer, LaBraM	No representation produced reliable unseen-patient preictal detection in the current four-subject study.
ning	LaBraM official pretrained backbone vs from-scratch SALT	LaBraM loaded 219 pretrained tensors but did not improve test preictal recall with the c bipolar-channel mapping.
mbalance	Weighted loss / controlled interictal downsampling	These mitigations did not yet provide stable held-out seizure-recording performance.
holding	Validation-derived thresholds	Thresholds must never be selected on the test recordings. High test AUC alone did not ensure usable alerts.

Current conclusion

This is an early feasibility study, not final PoMAS performance. The next proposed-model experiment is patient-dependent STFT spectrogram plus inter-channel correlation features with EfficientNet-B0 and an SVM risk classifier. Final claims require recording-level cross-validation across more subjects and event-level alert metrics.